



Next-Gen Credit Risk and Fraud Detection with Explainable AI

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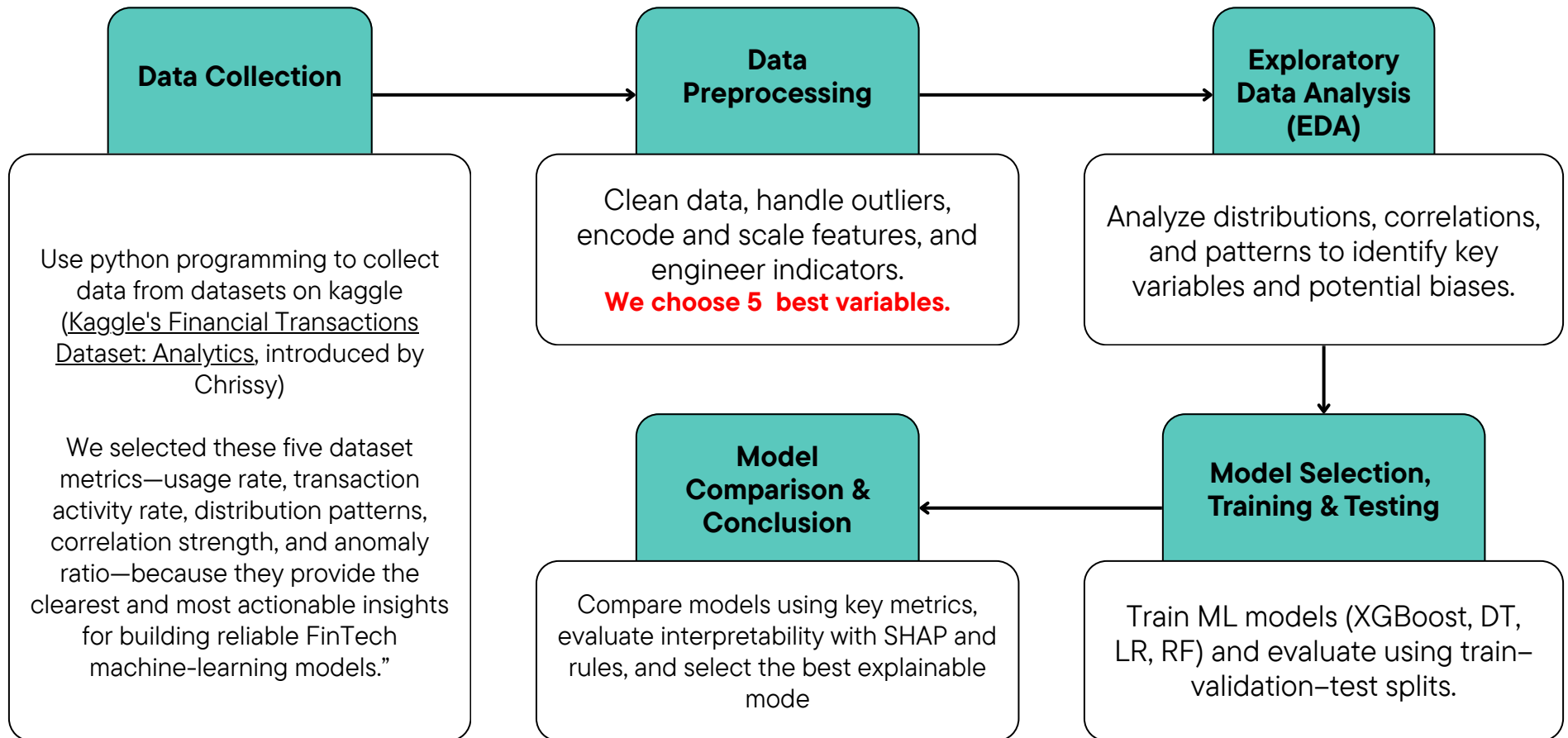


Motivation

- There is a need to redesign ML frameworks for consumer FinTech analytics with both accuracy and interpretability.
- The model should capture complex data patterns across users, cards, and transactions.
- It must remain transparent and trustworthy for users and regulators.
- This study proposes a hybrid framework combining:
 - XGBoost for strong predictive power.
 - Decision Tree and SHAP for clear, explainable insights.



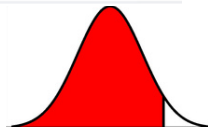
Overview Study Plan



Intro Datasets

- Teachers datasets on kaggle ([Kaggle's Financial Transactions Dataset: Analytics](#), introduced by Chrissy)
- Cards, Transactions, Users**

Item	Description	項目	說明	項目	說明
1 id	交易 ID	1 id	卡片 ID	1 客戶 ID	客戶的唯一識別碼
2 date	交易時間 (YYYY/MM/DD HH:MM)	2 client_id	客戶 ID	2 目前年齡	客戶當前的年齡
3 client_id	客戶 ID	3 card_brand	卡片品牌 (如 Visa, MasterCard)	3 預計退休年齡	客戶計劃退休的年齡
4 card_id	卡片 ID	4 card_type	卡片類型 (Debit / Credit)	4 出生年份	客戶的出生年份
5 amount	交易金額	5 card_number	卡號 (數字格式)	5 出生月份	客戶的出生月份
6 use_chip	交易方式 (Chip Transaction / Swipe Transaction / Online)	6 expires	到期日 (MM/YYYY)	6 性別	客戶的性別
7 merchant_id	商家 ID	7 cvv	安全碼	7 住址	客戶的居住地址
8 merchant_city	商家所在城市	8 has_chip	是否有晶片	8 住家緯度	客戶住址的緯度座標
9 merchant_state	商家所在州	9 num_cards_issued	客戶已發行的卡片數量	9 住家經度	客戶住址的經度座標
10 zip	商家郵遞區號	10 credit_limit	信用額度 (美元, 需轉為數字)	10 當地人均收入	客戶所在地區的平均人均收入
11 mcc	Merchant Category Code (商家類別碼)	11 acct_open_date	開卡日期 (MM/YYYY)	11 使用者年收入	客戶的年度總收入
12 errors	交易錯誤訊息	12 year_pin_last_changed	最近一次修改 PIN 的年份	12 總負債	客戶的總負債金額
		13 card_on_dark_web	卡片是否出現在暗網	13 信用評分	客戶的信用評分數值
				14 擁有的信用卡數	客戶持有的信用卡數量



Clean the data

- There datasets we use is **Cards, Transactions, Users**
- Remove or make up the missing values
- Combine three datasets
- Remove unused datas (cvv, card_on_dark_web...)

	id_txn	date	client_id_txn	card_id	amount	use_chip	merchant_id	...	total_debt	credit_score	num_credit_cards	hour	date_only	weekday	date_hour
0	7475327	2010-01-01 00:01:00	1556	2972	-77.00	Swipe Transaction	59935	...	110153.0	740	4	0	2010-01-01	4	2010-01-01
1	7475328	2010-01-01 00:02:00	561	4575	14.57	Swipe Transaction	67570	...	112139.0	834	5	0	2010-01-01	4	2010-01-01
2	7475329	2010-01-01 00:02:00	1129	102	80.00	Swipe Transaction	27092	...	36540.0	686	3	0	2010-01-01	4	2010-01-01
3	7475331	2010-01-01 00:05:00	430	2860	200.00	Swipe Transaction	27092	...	128676.0	685	5	0	2010-01-01	4	2010-01-01
4	7475332	2010-01-01 00:06:00	848	3915	46.41	Swipe Transaction	13051	...	96182.0	711	2	0	2010-01-01	4	2010-01-01
5	7475333	2010-01-01 00:07:00	1807	165	4.81	Swipe Transaction	20519	...	98613.0	828	5	0	2010-01-01	4	2010-01-01
6	7475334	2010-01-01 00:09:00	1556	2972	77.00	Swipe Transaction	59935	...	110153.0	740	4	0	2010-01-01	4	2010-01-01
7	7475335	2010-01-01 00:14:00	1684	2140	26.46	Online Transaction	39021	...	108313.0	782	5	0	2010-01-01	4	2010-01-01
8	7475336	2010-01-01 00:21:00	335	5131	261.58	Online Transaction	50292	...	66565.0	688	3	0	2010-01-01	4	2010-01-01
9	7475337	2010-01-01 00:21:00	351	1112	10.74	Swipe Transaction	3864	...	375.0	807	6	0	2010-01-01	4	2010-01-01



Define Credit Fraud

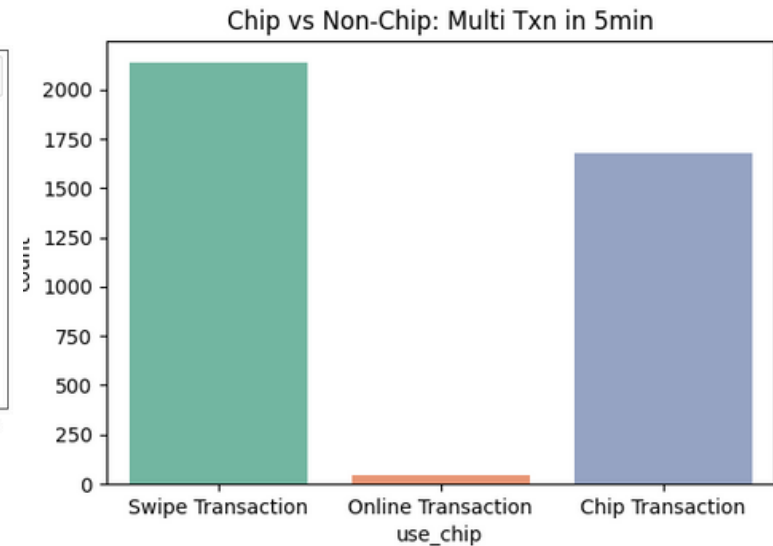
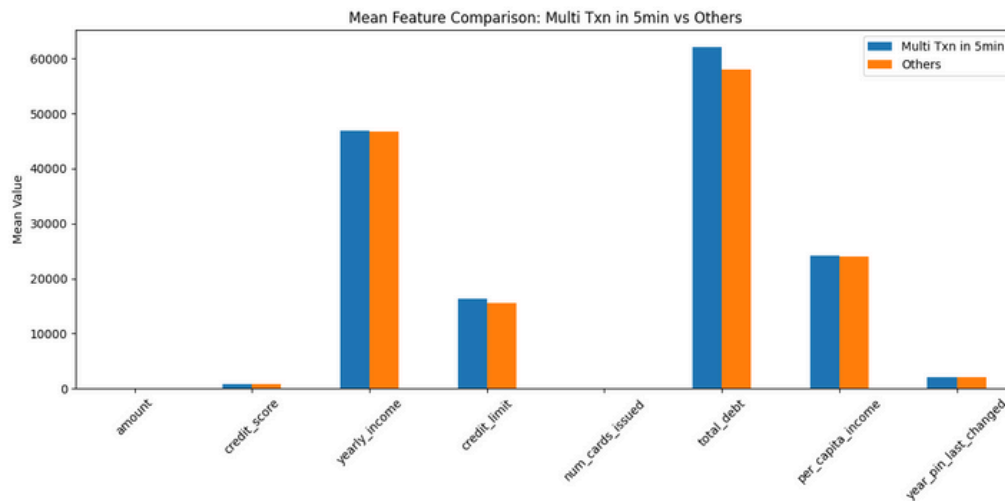
- To define what situations conform to fraud.
- Common situations
 - Low credit score
 - **Late-night transactions ✓**
 - **Multiple transactions in a short period ✓**
 - Spending amount significantly different from usual
 - Spending location too far from home
 - Don't change card pin for long time



EDA Chart

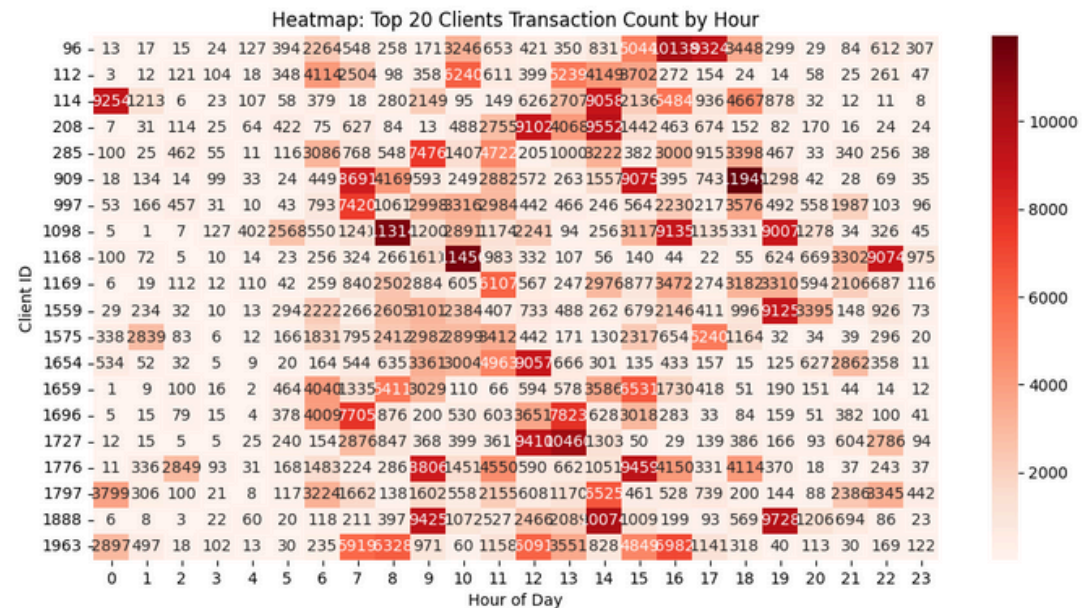
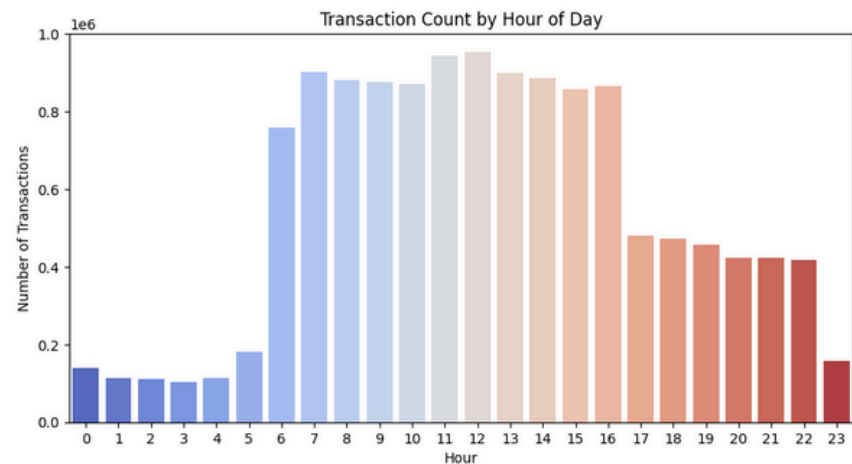
- **Frequent transactions in a short period of time (5min)**
- Notated these datas in Trans_Data
- Measure the difference in these bias data

五分鐘內多筆交易 (>3筆) 共有 3862 筆，佔全部交易 0.03%



EDA Chart

- About transactions Time
- Observe numerous amount at strange time
 - Client 114
 - Client 1168



Conclusion

- Transaction Distribution
 - Trading is concentrated between 6 AM and 4 PM.
 - The least activity occurs from 11 PM to 5 AM.
- The risk of using a card by swiping is the higher than chip.
- High transaction clients compared to low transaction clients.
 - higher yearly income
 - higher total debet
- Frequent transactions data compared to others.
 - higher total debet
 - higher credit limit





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